Introduction to neuromorphic engineering and design approaches

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ISCAS Tutorial
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The zoo of neuromorphic hardware

An overview of neuromorphic platforms

Why such a diversity?

Subthreshold analog (mixed-signal)

Biological-time brain emulation and basic research

ROLLS (UZH/ETHz) — *DYNAPs (UZH/ETHz)* — *NeuroGrid/Braindrop (Stanford)*

Software

Low-cost simulation: neuromorphic (slow), neural networks (fast)

CPU / GPU

Dedicated/distributed sim.

Simulation acceleration for neuroscience and neural networks

FPGA — *SpiNNaker 1/2 (Manchester, Dresden)*

Above-threshold analog (mixed-signal)

Neuroscience simulation acceleration

BrainScaleS 1/2 (Heidelberg)

Large-scale full-custom digital designs

Cognitive computing

TrueNorth (IBM) — *Loihi (Intel)*

Small-scale full-custom digital designs

open source hardware

Edge computing

(exploration platform) — ***(accelerator)***

ODIN (UCLouvain) — *ReckOn (UZH/ETHz)*

See also:

	[Seo, CICC'11]
	[Knag, JSSC'15]
MorphIC	[Frenkel, Trans. BioCAS'19]
	[Park, ISSCC'19]
SPOON	[Frenkel, ISCAS'20]
μBrain	[Stuijt, Front. Neur'21]
SENeCA	[Yousefzadeh, AICAS'22]

Can we ignore neuroscience insight toward intelligent systems?

The case of neuromorphic computing

Can we find a middle gro

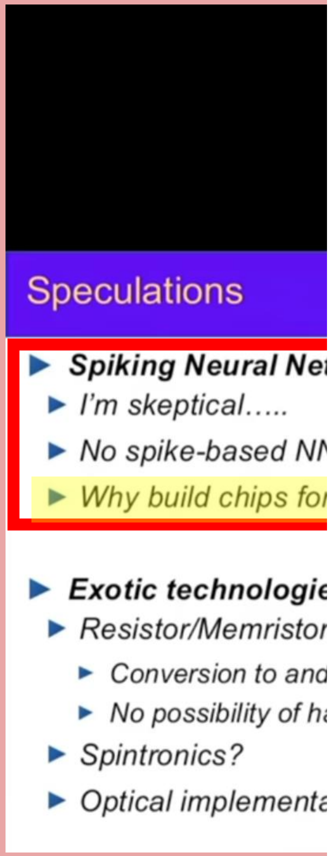
"The brain is the gold standard for efficiency"

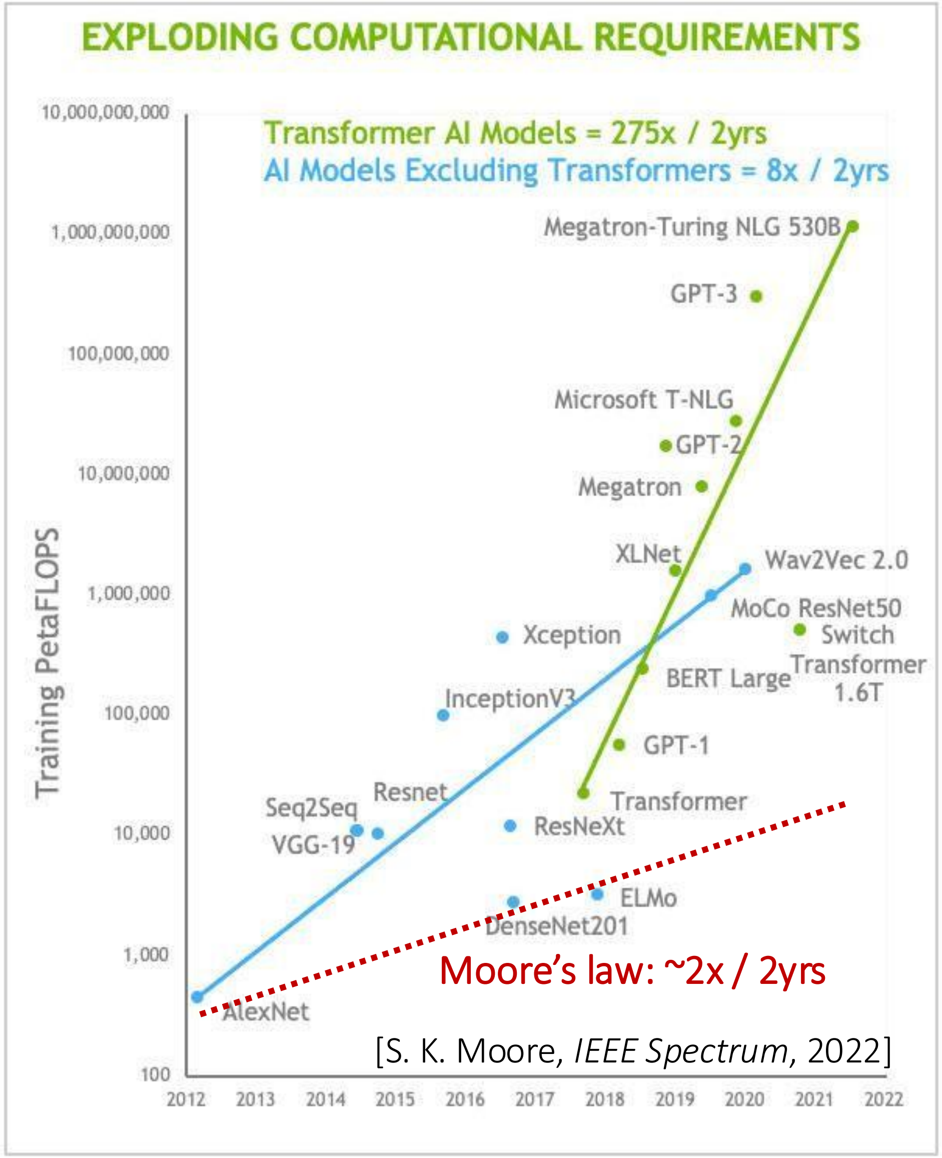
"AI is not sustainable"

"Brain-inspired = low-power"

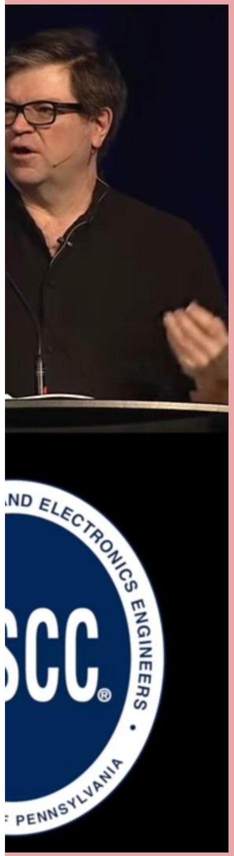
Timely →

"Key technology!"





[S. K. Moore, *IEEE Spectrum*, 2022]



Outline

Let's dive into neuromorphic hardware design principles and key tradeoffs!

- Analog vs. digital
- Fully parallel vs. time-multiplexed
- Information locality
- Bottom-up vs. top-down

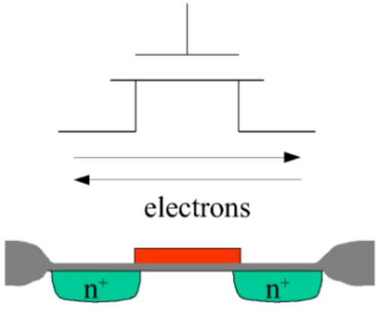
LIF neuron:

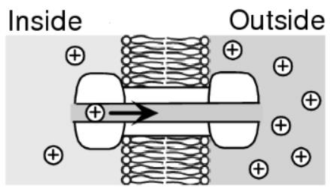
$$V[t+1] = \underbrace{\beta V[t]}_{\text{Can be made event-driven too!}} + \underbrace{W \cdot X[t]}_{\text{Should be event-driven}} \quad \text{(costly!)}$$

Remember the previous update time.

Simulation → Phenomenological modeling

Analog vs. digital





Biophysical behavior

emulation → Analog

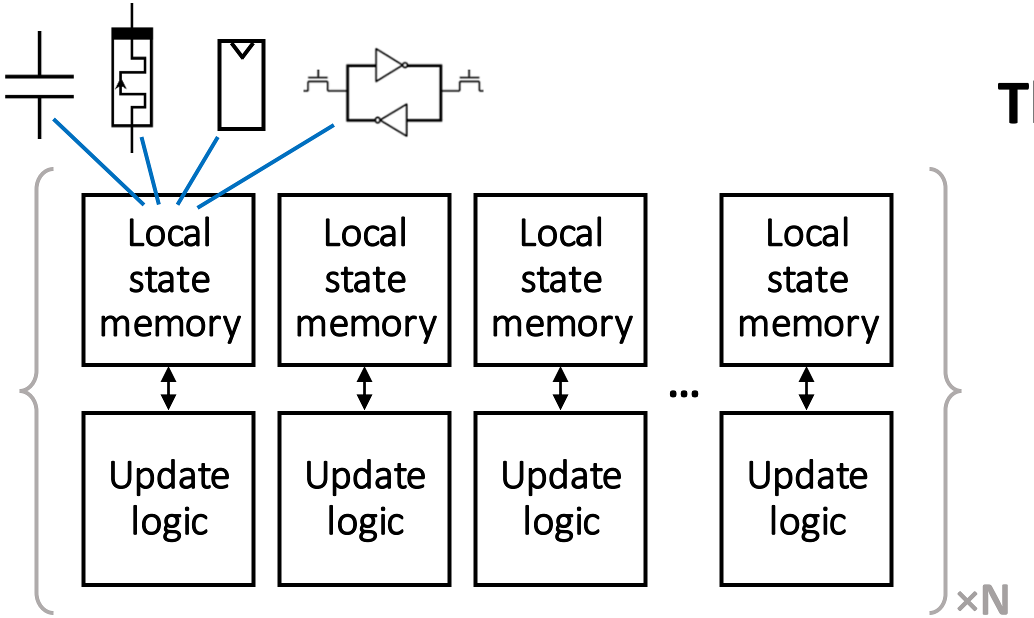
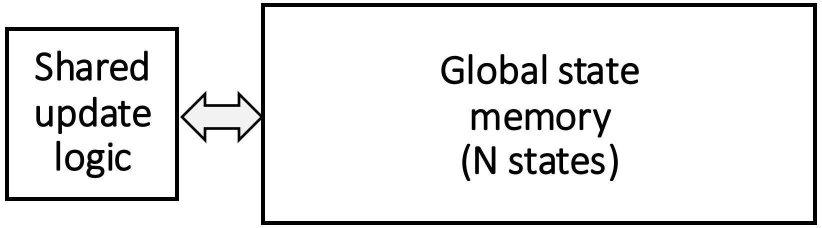
~~simulation~~ → Digital

Digital → phenomenological modeling → Analog

	Analog	Digital
Biophysical accuracy / Area	Good	Bad
Design time	High	Low
Technology scaling	Low potential	High potential
Noise, mismatch, PVT	Bug or feature?	Low sensitivity
Go-to solution for:	Low power (/!\ at system level!), complex dynamics "for free"	Flexibility (design space, programmability), design time vs. efficiency

That's why open source hardware is mostly digital for now, but new exciting open-source mixed-signal designs are coming!

Fully parallel vs. time-multiplexed architectures

	Diagram	Throughput	Area	Power
Fully parallel architecture		✓		~ Overhead in static power
Time-multiplexed architecture			✓	~ Overhead in dynamic power

wooclap x4

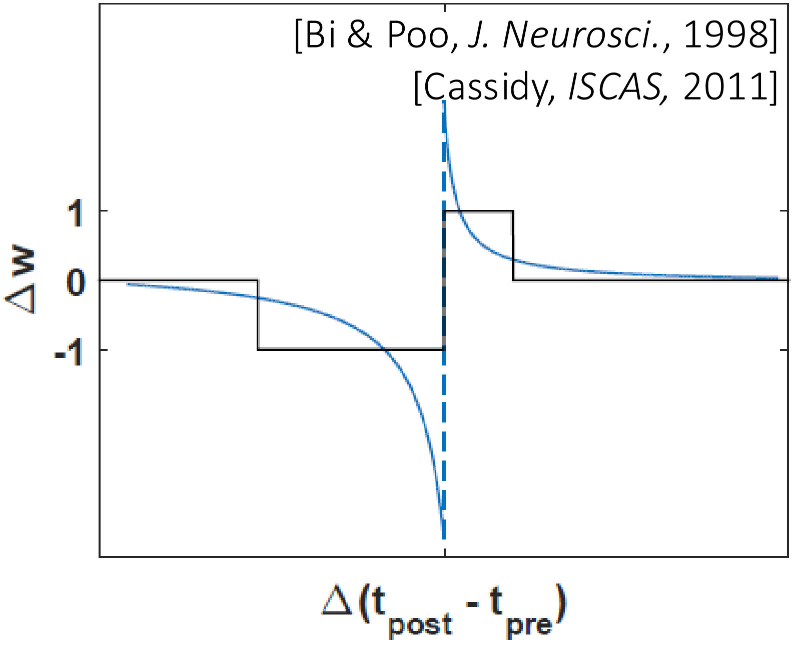
Now, can you see why digital PDE solvers are so costly?

[Frenkel, *Proc. IEEE*, 2023]

Information locality

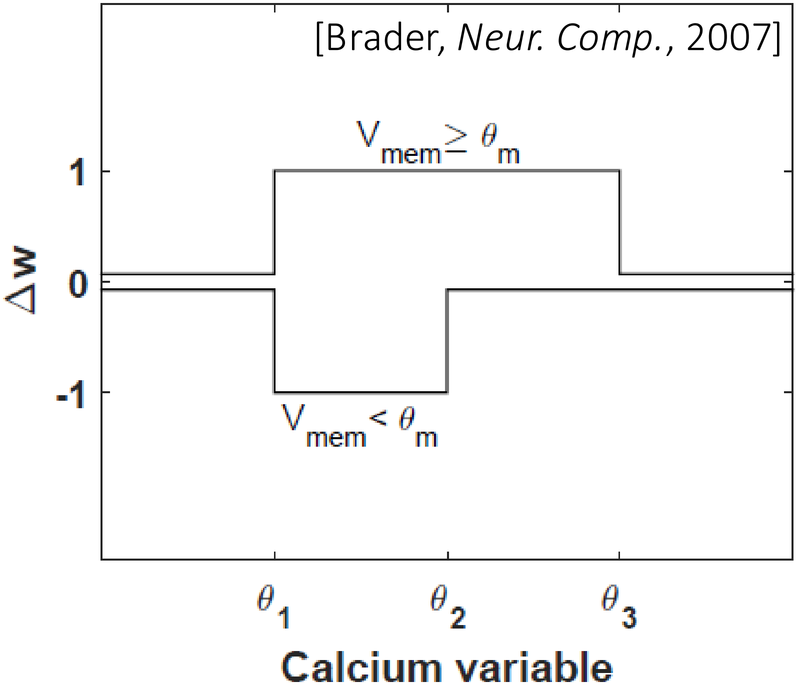
Illustrated with "local" two-factor spike-based learning rules

Spike-timing-dependent plasticity (STDP)

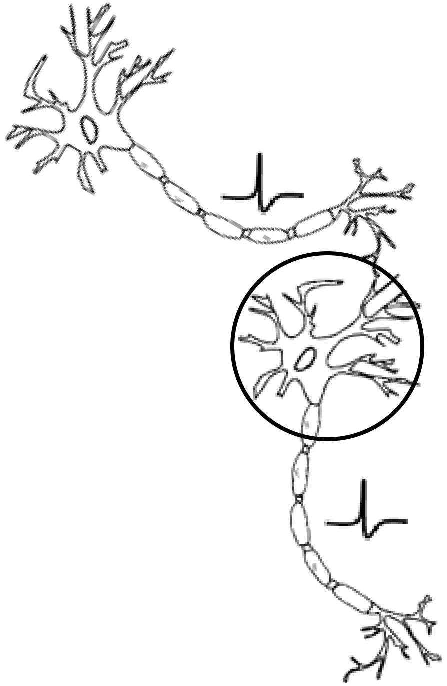


✔ **Local**

Spike-dependent synaptic plasticity (SDSP)



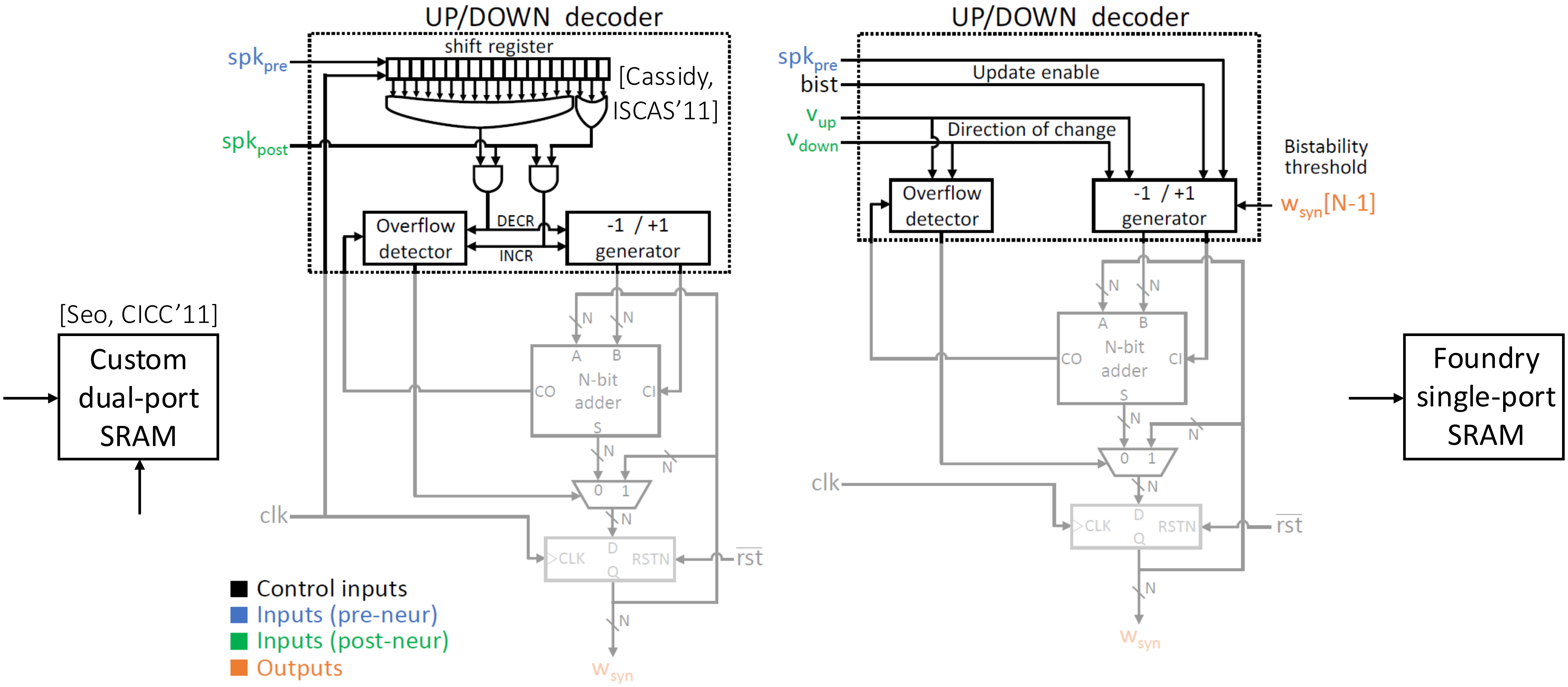
✔ **Local**



Information locality

Illustrated with "local" two-factor spike-based learning rules

Key challenge – Fan-in = 100-10000 synapses/neuron



STDP SDSP

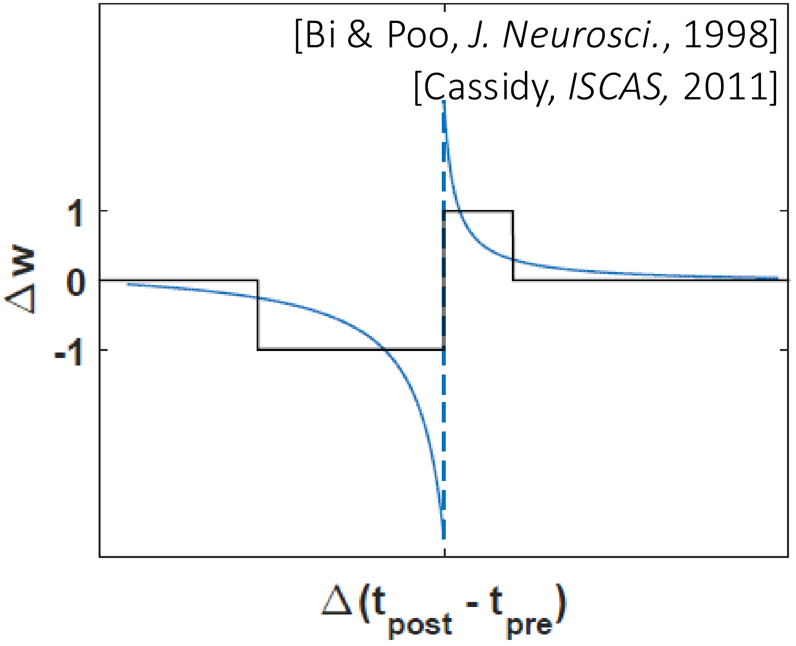
[Frenkel, *ISCAS*, 2017]

Information locality

Illustrated with "local" two-factor spike-based learning rules

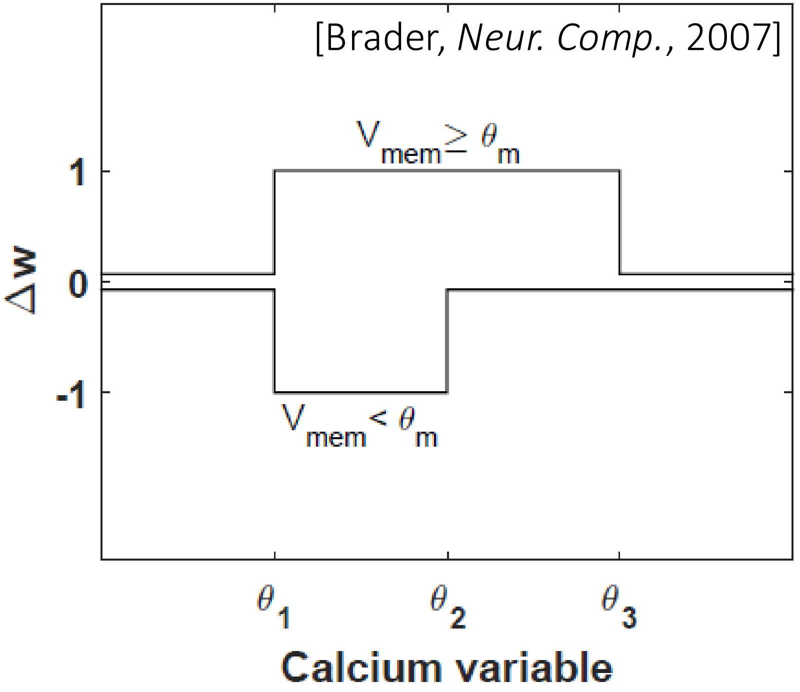
"Locality is all you need"

Spike-timing-dependent plasticity (STDP)



✔ **Local** in space
✘ **Non-local** in time

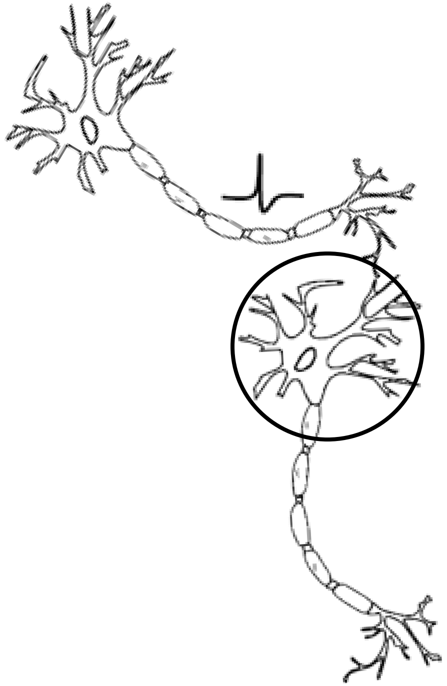
Spike-dependent synaptic plasticity (SDSP)



✔ **Local** in space
✔ **Local** in time

$\underbrace{\hspace{6em}}$

Huge savings in silicon



[Clopath and Gerstner, *Front. Syn. Neuro.*, 2010]

[Frenkel, *TBioCAS*, 2019]

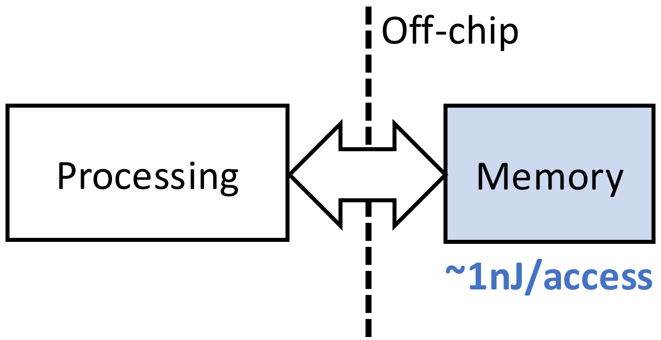
Why does locality matter for hardware?

Convergence

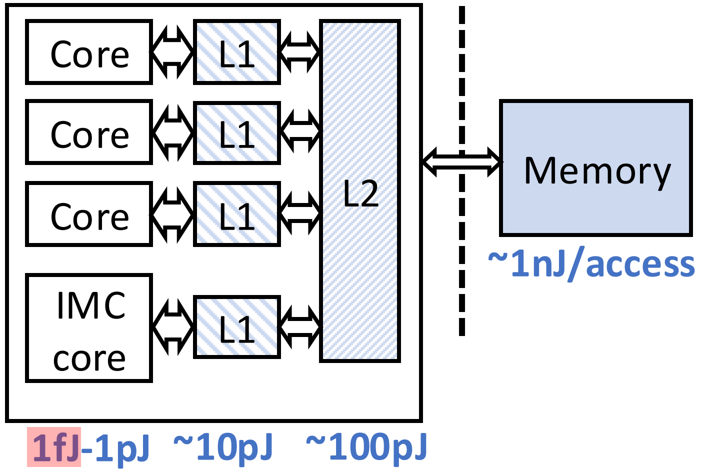
Engineering

(Neuro)science

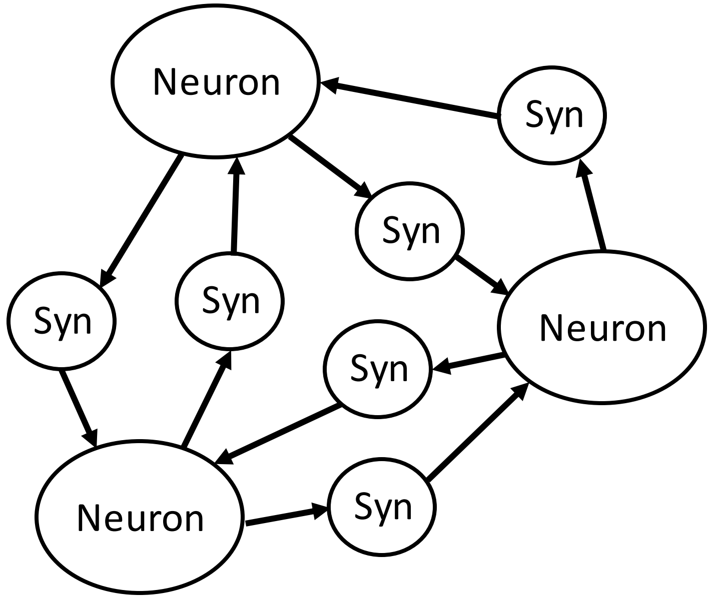
Von-Neumann architecture



Optimized von-Neumann architecture



Bio-inspired architecture



Memory!

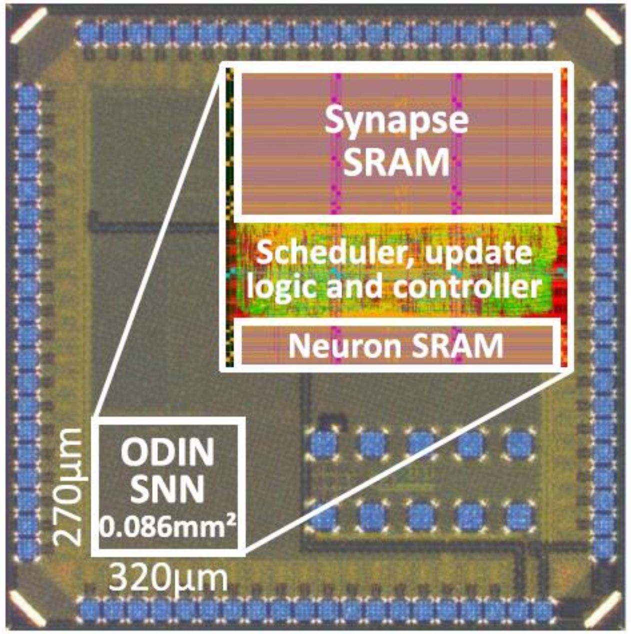
Backprop-based training ***(Global)***

Spike-time-based training ***(Local)***

Ensuring strict locality in space and time breaks records

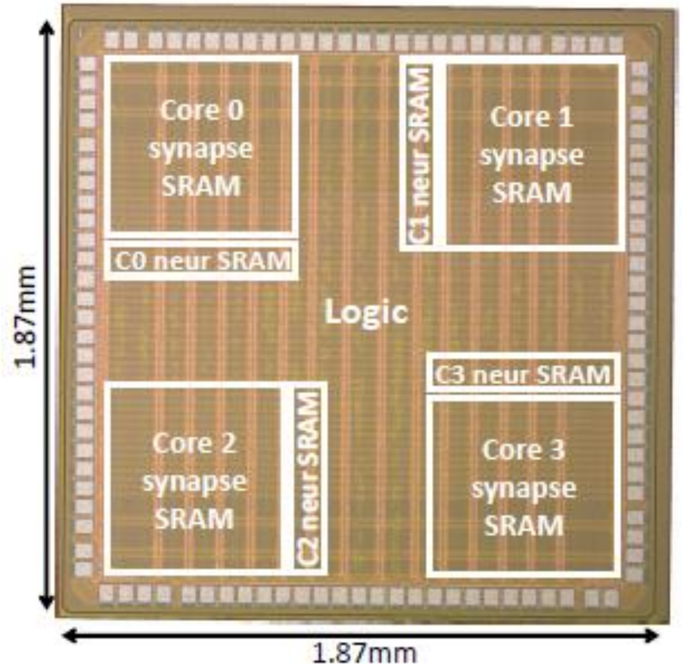
ODIN (single-core)

[Frenkel, TBioCAS'19a]



MorphIC (quad-core)

[Frenkel, TBioCAS'19b]

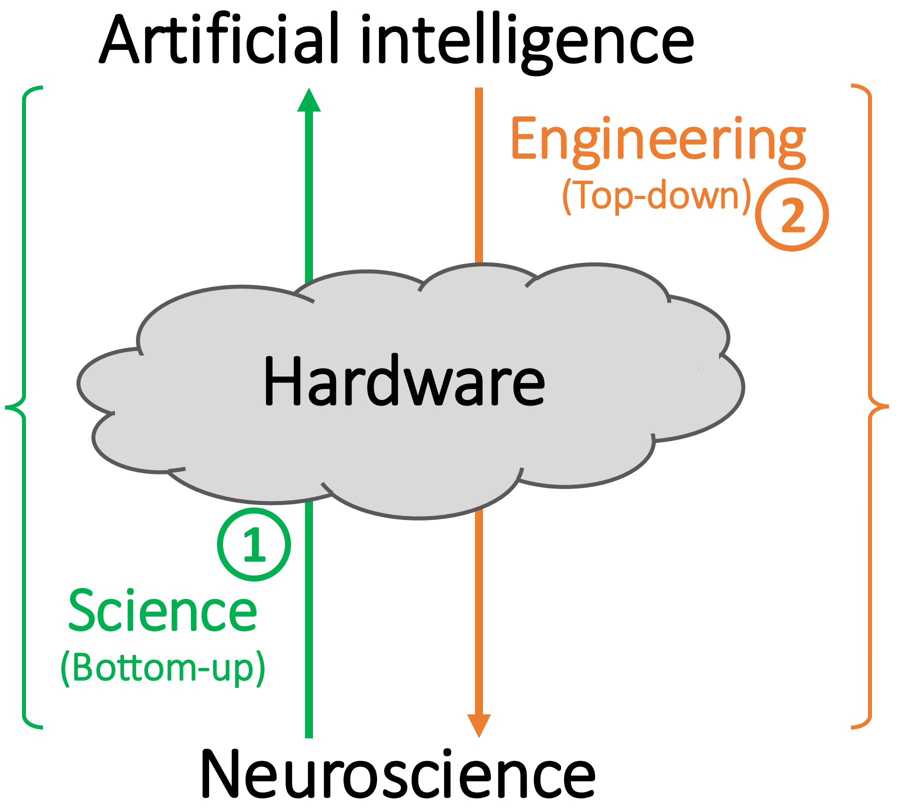


open source hardware

- ✔ Record synaptic density (and includes plasticity)
- ✔ Energy efficiency competitive with mixed-signal designs

...but quite painful to exploit!

Let’s zoom back a bit…



Most of today’s neuromorphic engineering

- ✔ Analysis-by-synthesis
- ✘ Difficult to scale efficiently to real-world problems

Today’s AI hardware accelerators

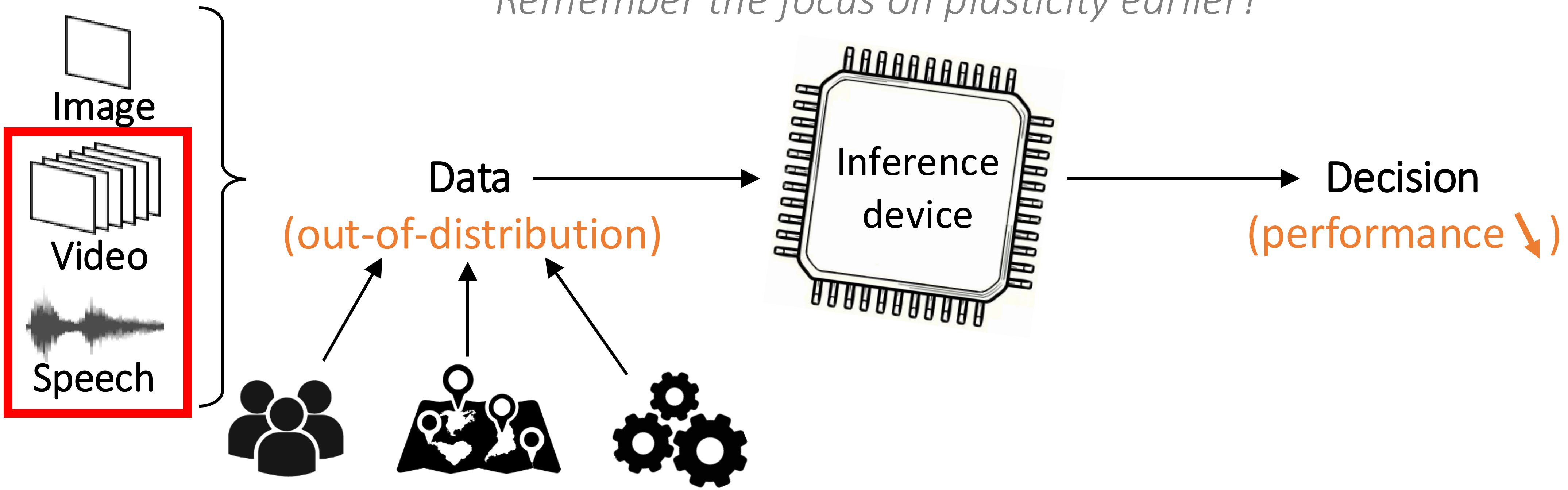
- ✔ Starts from working solutions to real-world problems
- ✘ Which “salt & pepper” from neuroscience?

Don’t pick a side, make the best of both through co-design!

Neuromorphic intelligence:

1) Pick the use case – On-device learning!

Remember the focus on plasticity earlier?



Different users, environments, task requirements

More training data before deployment?

Issues: cost, robustness, flexibility

Data exchange with the cloud?

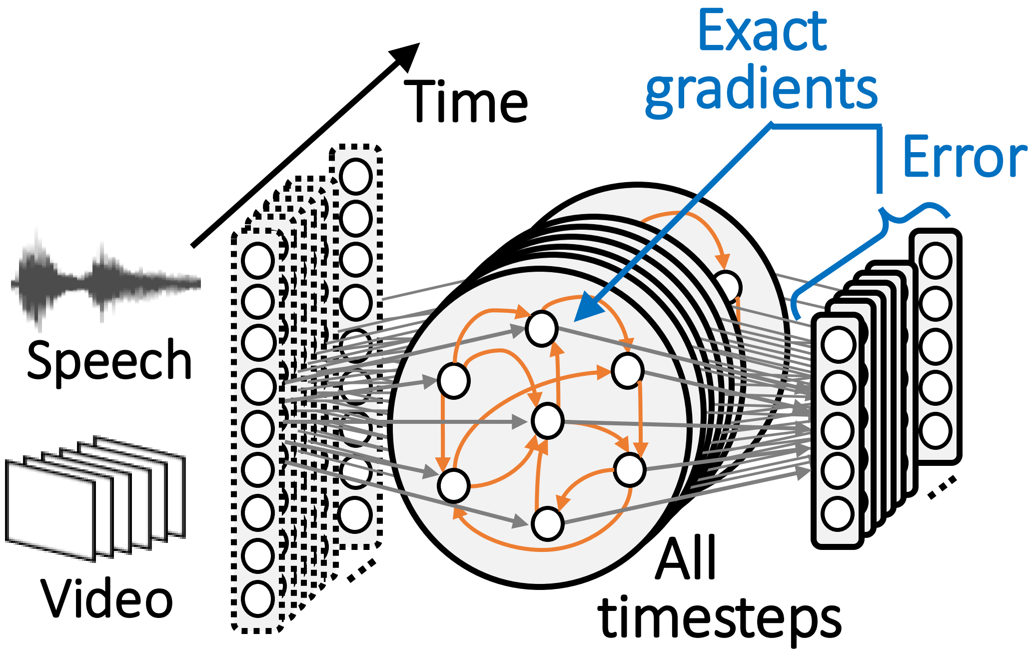
Issues: power budget, privacy

On-chip training
(end-to-end)

2) Select the (ML-informed) starting point

From backpropagation through time to bio-plausible learning

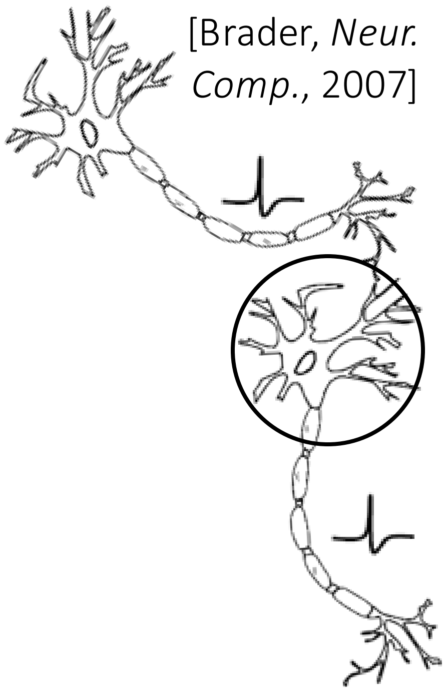
Backprop through time (BPTT, backward)



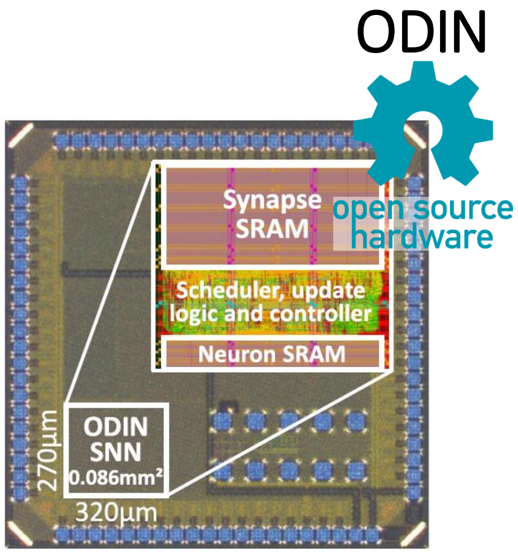
$$\frac{\mathrm{dE}}{\mathrm{dW_{ij}}}$$

Obviously, the brain doesn't do that!

Spike-timing-dependent plasticity



[Brader, *Neur. Comp.*, 2007]

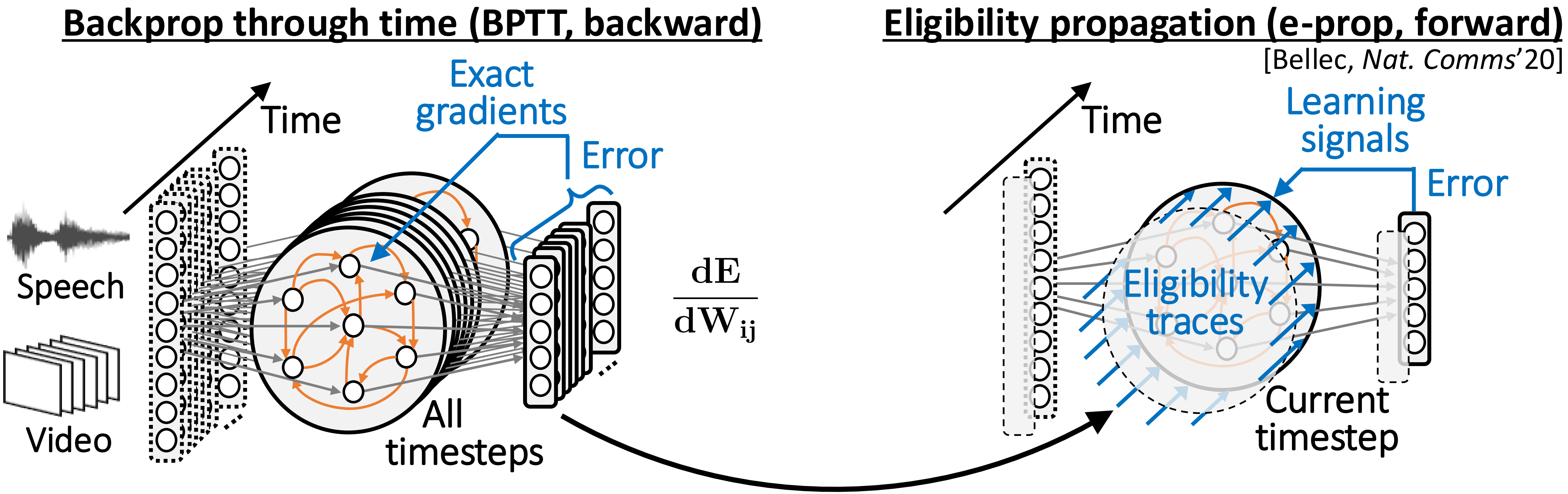


[Frenkel, TBioCAS, 2019]

- ✔ **Local** in space
- ✘ ~~**Local**~~ in time
- ✘ Doesn't work beyond MNIST…

2) Select the (ML-informed) starting point

From backpropagation through time to bio-plausible learning



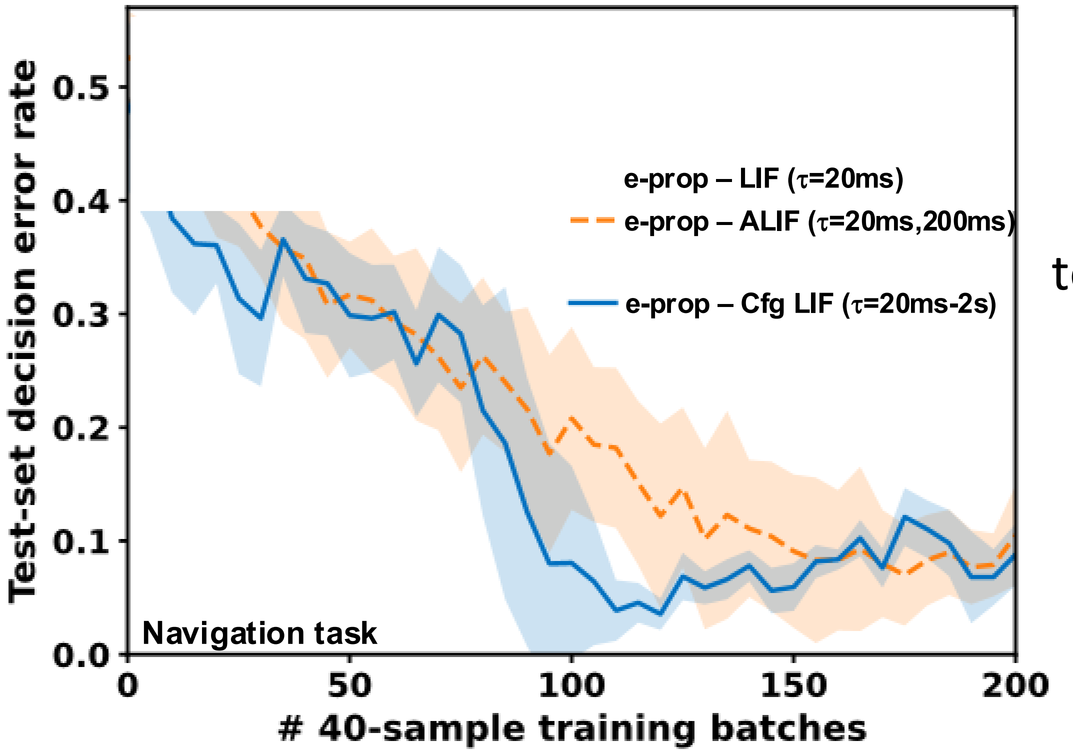
Obviously, the brain doesn't do that!

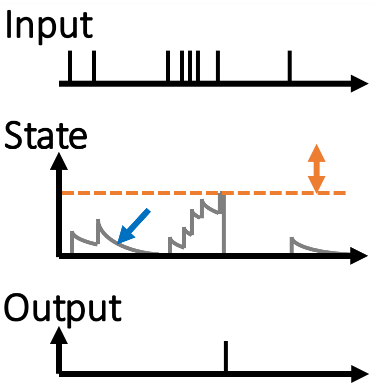
Key concept: space and time locality (again!)

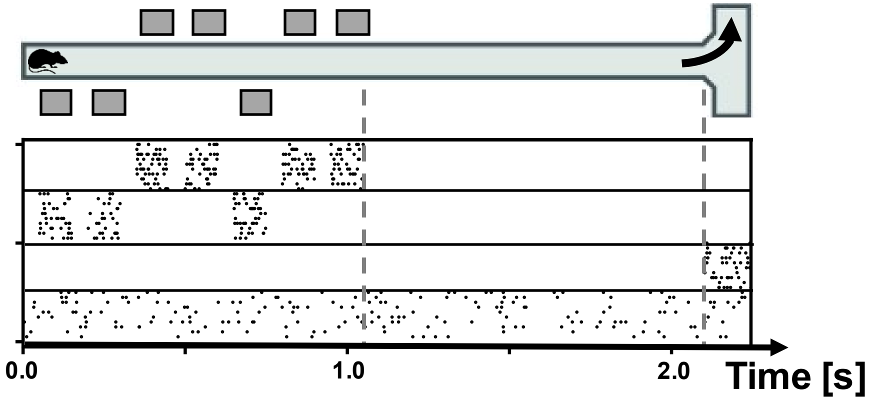
3) Pick the right feature set… and navigate the interplay of NeuroAI!

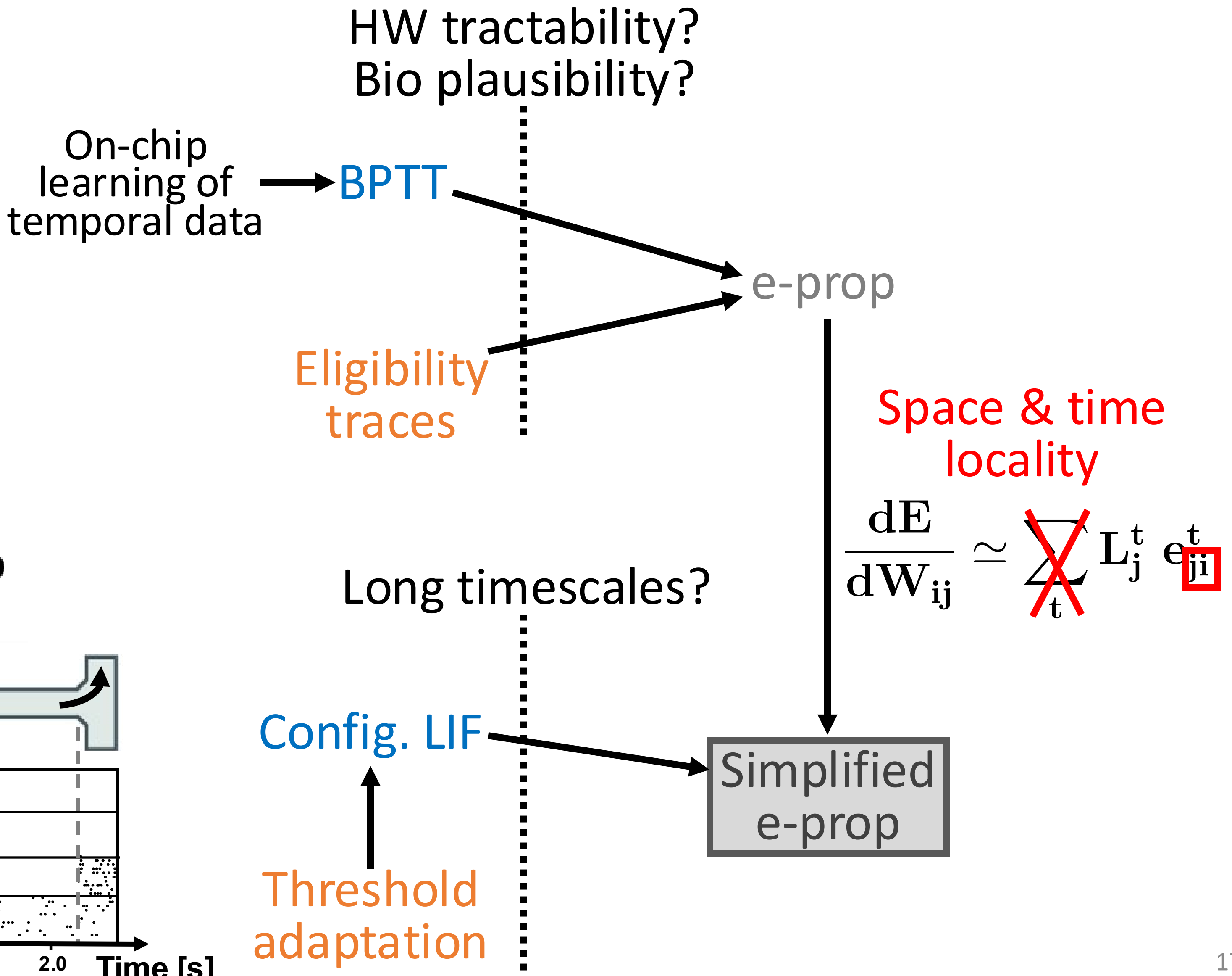
Neuron model selection… driven by the application requirements!

3) Pick the right feature set… and navigate the interplay of NeuroAI!



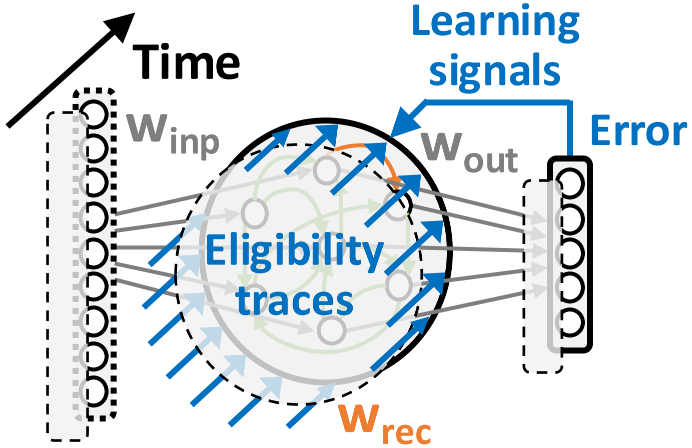






4) Enforce space and time locality

Key steps to minimize memory requirements



$$\frac{dE}{dW_{ij}} \simeq \cancel{\sum_t} L_j^t \, e_{\boxed{ji}}^t$$

Memory overhead scales with #synapses in $O(N^2)$

Local decoupling of space and time:

- **Pre-synaptic term:** activity low-pass filtering
- **Post-synaptic term:** surrogate derivative of the spiking activation function

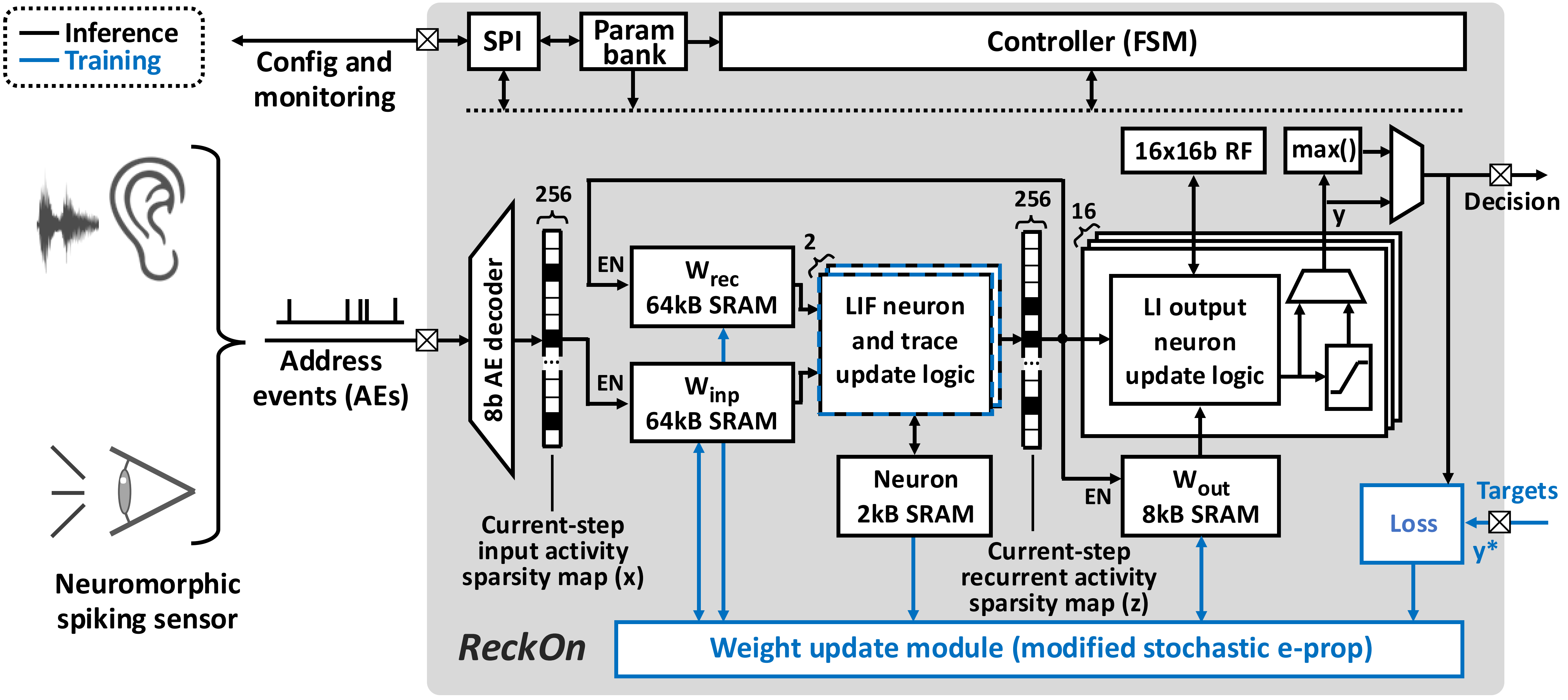
Scales with #neurons in O(N)

Only 1% memory overhead!

Stochastic weight updates allow reducing weight resolution to 8 bits

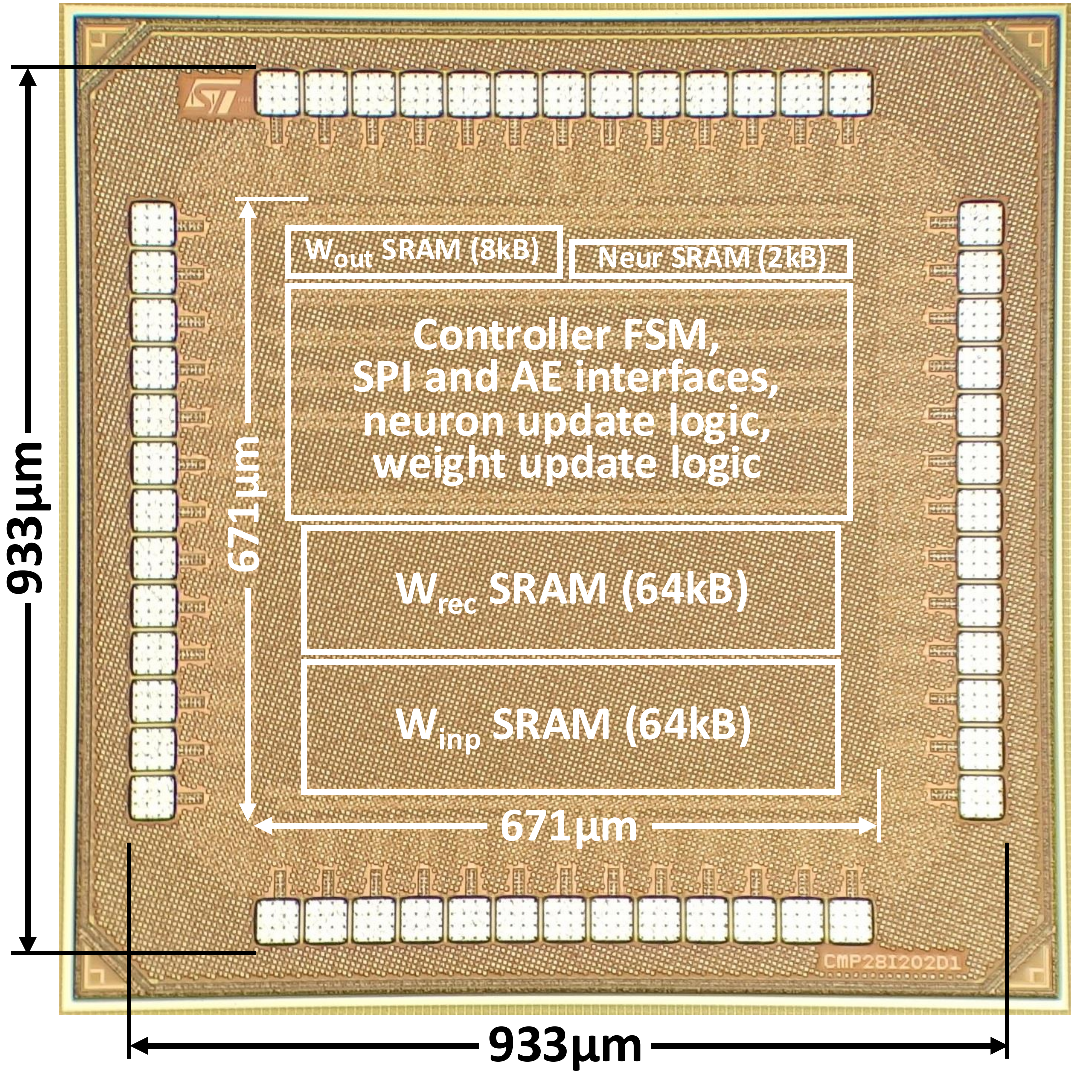
The ReckOn neuromorphic chip – Architecture

Same recipe as for ODIN: time multiplexing, no PDE solver, space and time locality



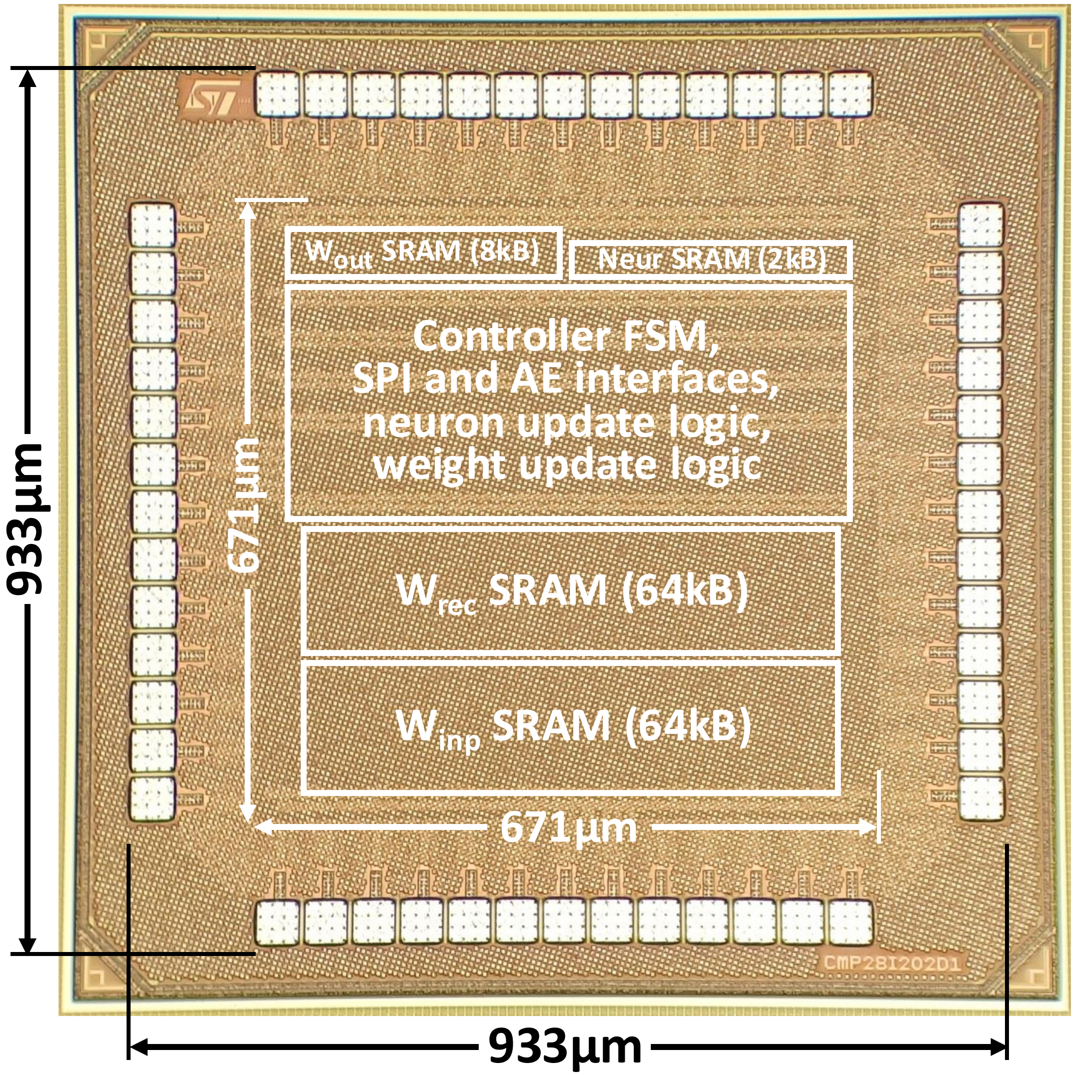
[Frenkel, *ISSCC*, 2022]

The ReckOn neuromorphic chip – Microphotograph and summary

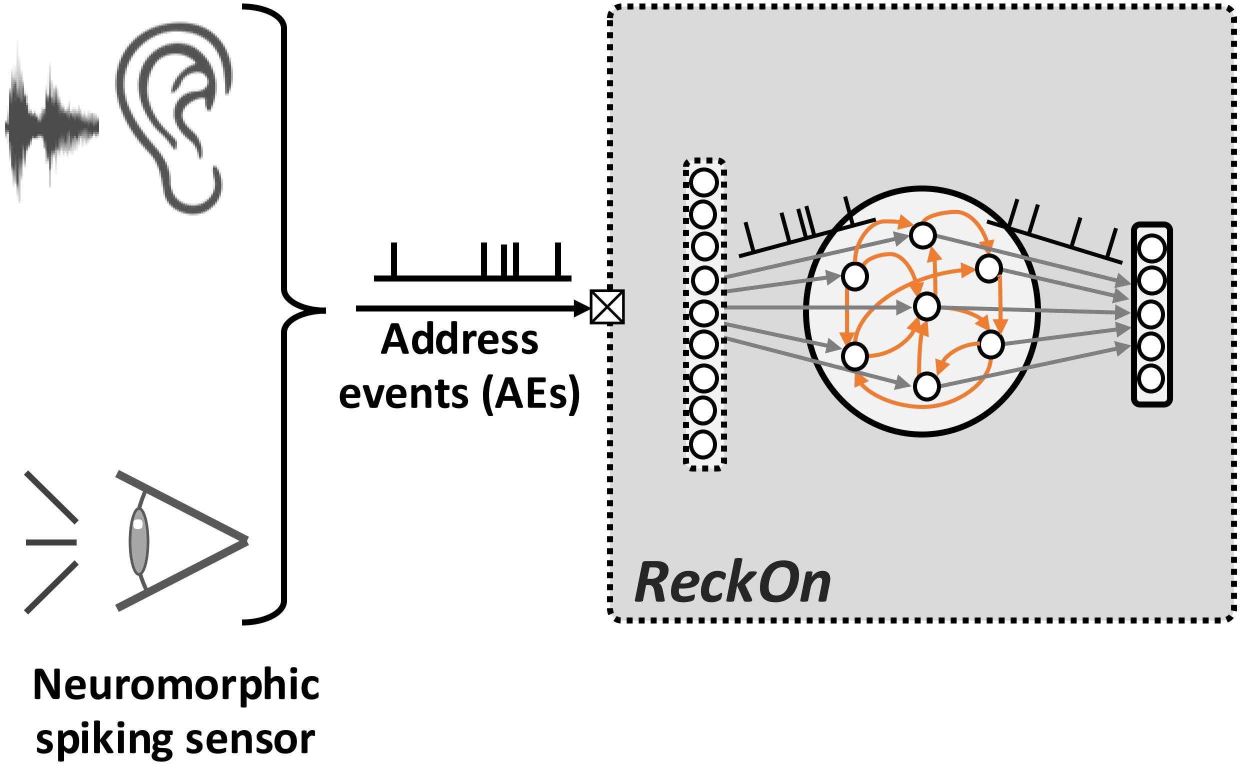


Technology	28nm FDSOI CMOS
Core size	0.67×0.67 mm² **0.45mm²**
Die size	0.93×0.93 mm²
SRAM	138kB **+ 0kB ext. DRAM!**
Network	Spiking RNN
Training timespan	Max. 32k steps

The ReckOn neuromorphic chip – Key advantage of using spikes



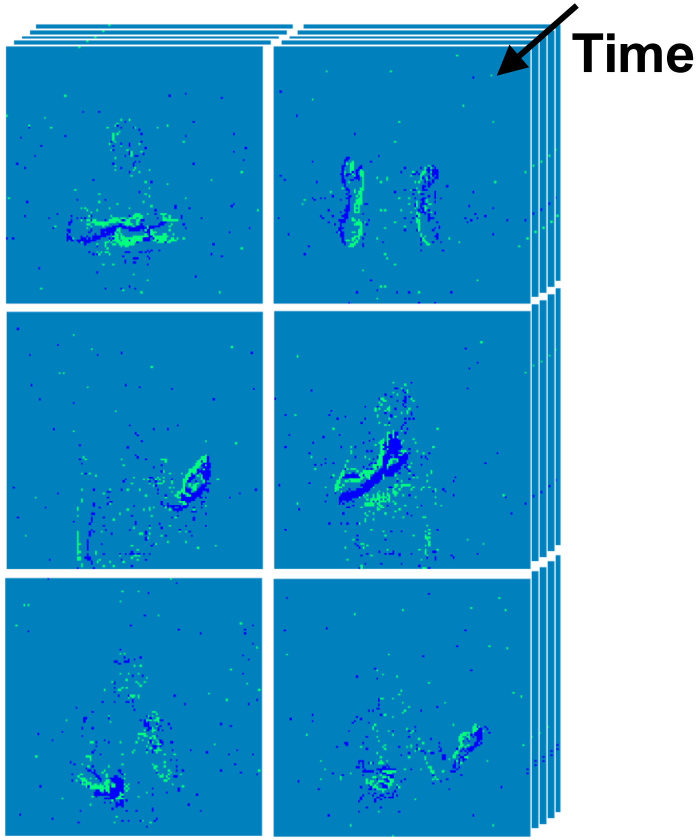
- Event-driven computation
- Sensor-agnostic raw-data processing
- Task-agnostic processing and learning



The ReckOn neuromorphic chip – Benchmarking

Vision

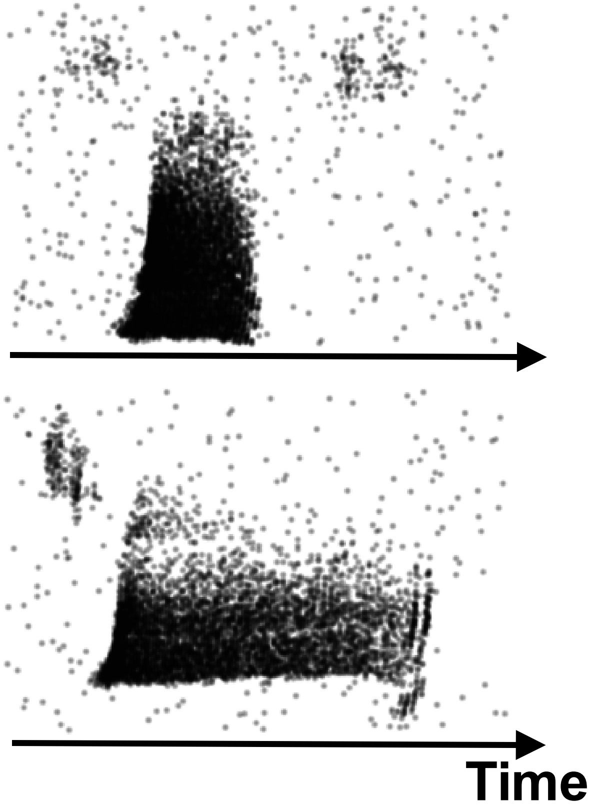
Gesture recognition
(DVS Gestures dataset)



Accuracy: 87.3% (28µW @0.5V)

Audition

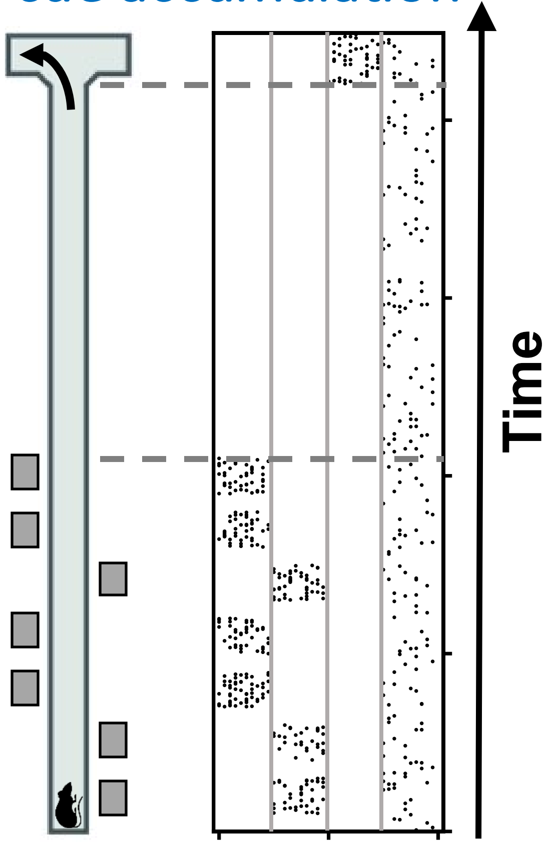
Keyword spotting
(Spiking Heidelberg Digits dataset)



Accuracy: 90.7% (46µW @0.5V)

Navigation

Delayed-supervision
cue accumulation



Accuracy: 96.4% (14µW @0.5V)